

TIKTOK VIRALITY ANALYSIS



IMC463 Machine Learning Final Project

Group 4

Alina, Ina, Yilin, Yuki



AGENDA

Business & Research Question



Descriptive Statistics



Ridge / Lasso for Variable Selection



OLS Model: Pre & Post Lasso



Regression Tree / Random Forest



Results & Marketing Implications



Limitations & Future Research



Research Question:

How do **emotional expressions in video comments** affect virality, measured by **share count**, on TikTok?



- 59% of people report that they frequently share online content with others. (Allsop, Bassett, and Hoskins 2007)
- Articles of the New York Times that evoke high-arousal positive (awe) or negative (anger or anxiety) emotions is more viral. (Berger & Milkman. 2012)

KEY VARIABLES

Dependent Variable: Engagement Metric

log_Share.Count

- Shares require active endorsement and spread content beyond the original audience
- Likes can be passive and views can be algorithm-driven.

Independent Variables: Comment-Based Features

log_mean_comment_disapproval	log_mean_comment_approval
log_mean_comment_disgust	log_mean_comment_admiration
log_mean_comment_anger	

- Emotional aspects of content may affect whether it is shared (Heath, Bell, and Sternberg 2001).
- **What we want to explore:** which is more likely to be shared, positive or negative content?

CONTROL VARIABLES

Selected control variables include **Creator-Related** & **Video Content Features**. **Demographic data** and **Visual Attributes & Quality** can capture the initial perceived impression of the video content, while **creator profile metrics** reflected the account's existing status and potential performance.

<u>Variable</u>	<u>Control Rationale</u>
Age	Viewer engagement may differ by perceived age due to age-related biases or relatability.
Gender.deepface	Gender may influence audience engagement and algorithmic visibility
Race	Race may affect viewer perception.
log_Follower.Count	Larger followings typically lead to higher engagement due to greater reach.
log_Likes.sum.Count	Reflects long-term popularity or account reputation.

<u>Variable</u>	<u>Control Rationale</u>
log_Following.Count	May reflect reciprocal engagement norms
Verified.Status	Verified accounts may receive algorithmic advantages and more user trust
Image_quality	Higher video quality may leads to better viewer experience.
Brightness	Brighter videos tend to affects mood and visibility.
Sharpness	High visual-clarity related to better production value and viewer retention

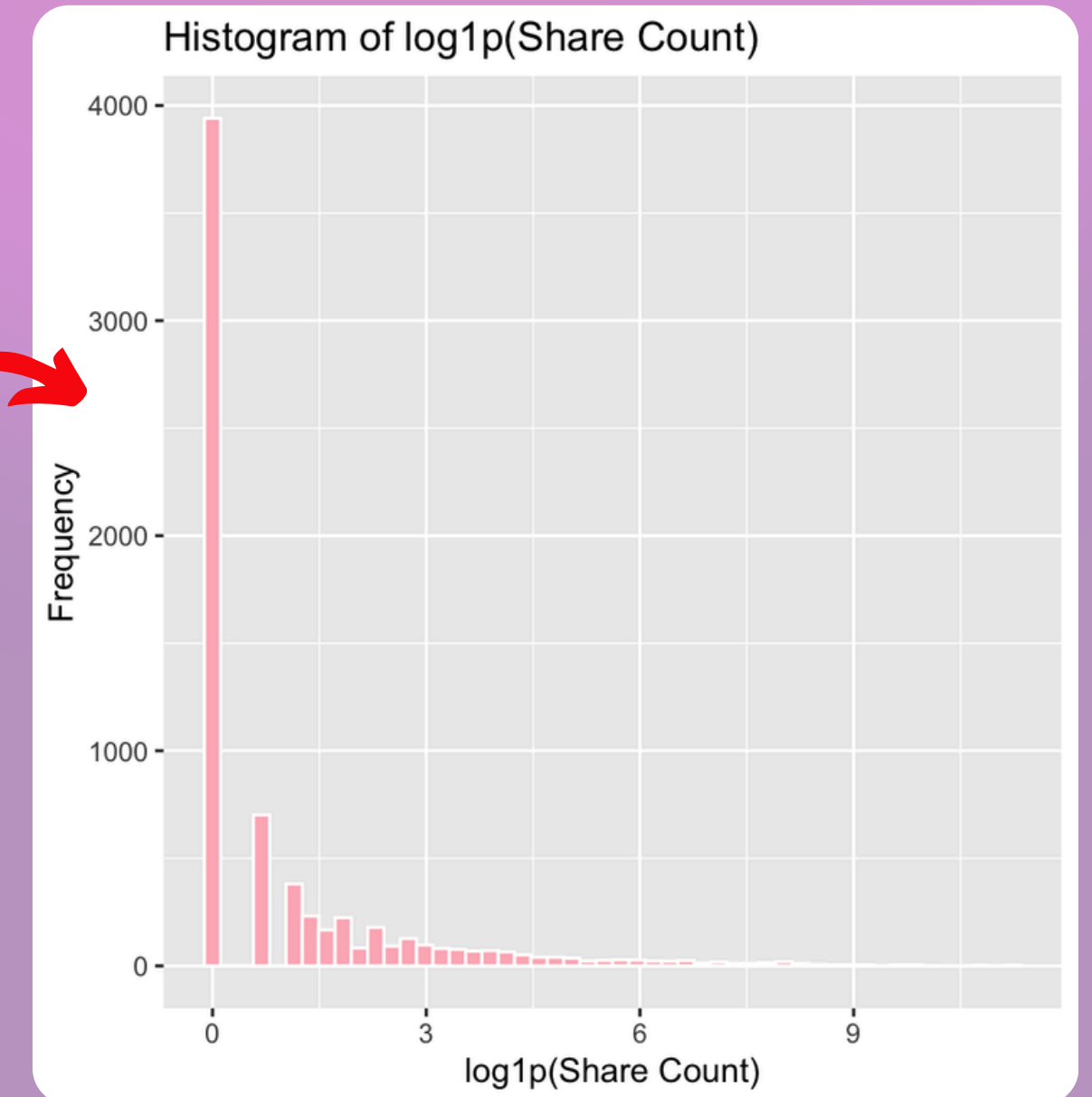
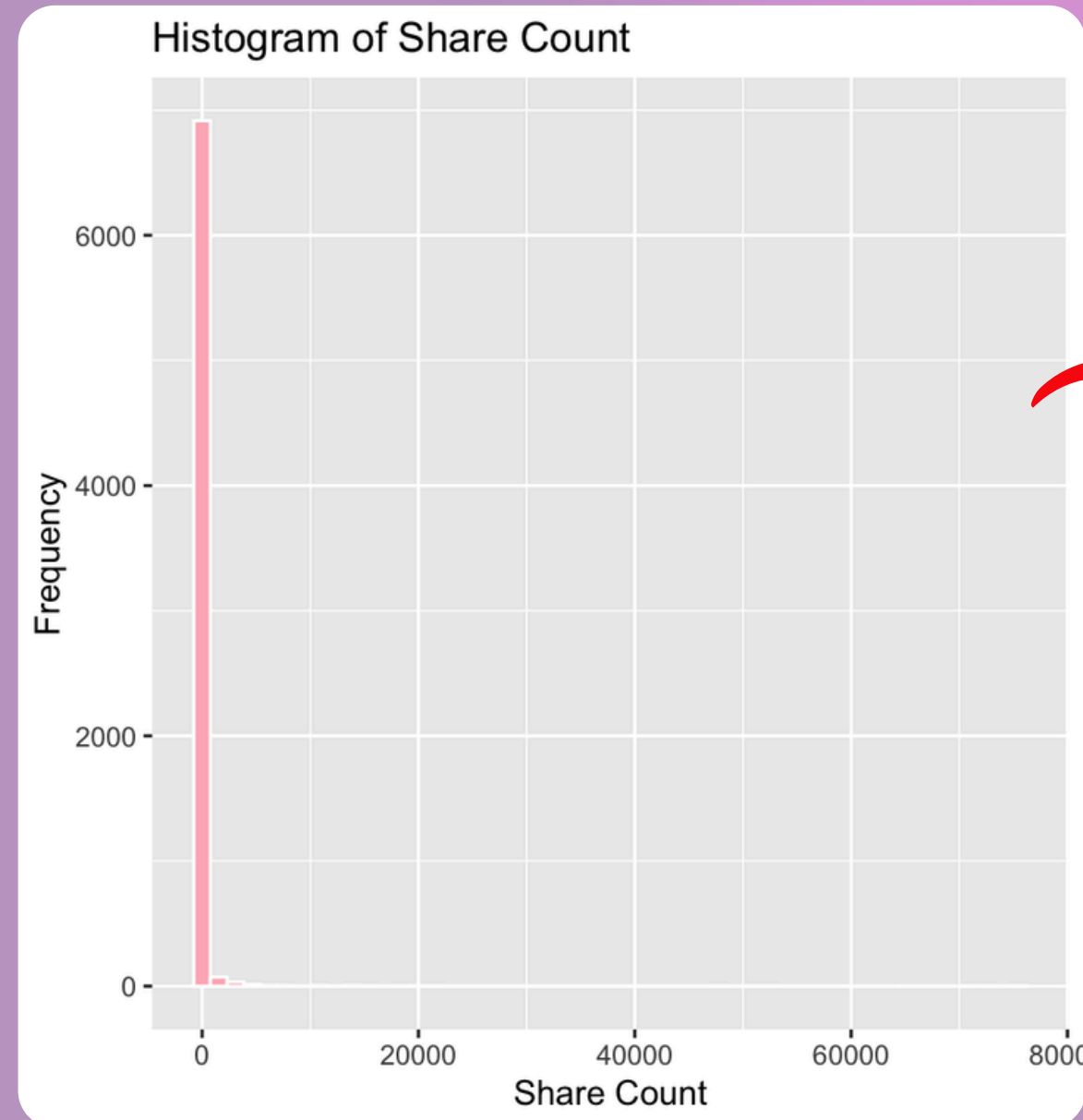
DESCRIPTIVE STATISTICS

Log-transforming

- Share.Count

ensures better model fit,
reduce extreme right-
skewness (skewness > 28).

	mean	sd	skew
Share.Count	142.54	1990.73	28.72



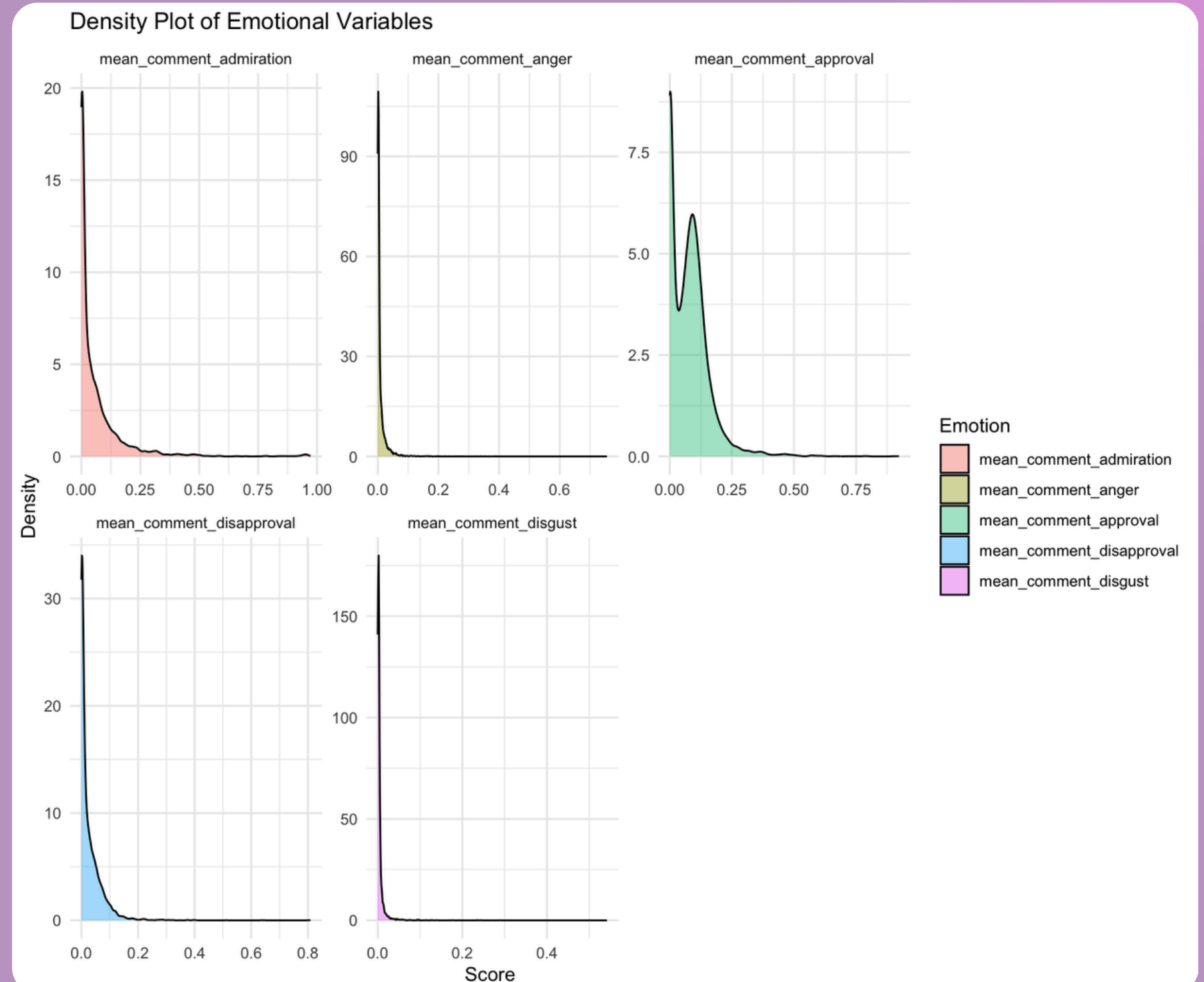
DESCRIPTIVE STATISTICS

Log-transforming:

- mean_comment_disapproval
- mean_comment_approval
- mean_comment_admiration
- mean_comment_disgust
- mean_comment_anger

reduces extreme right-skewness with log transformation.

	mean	sd	skew
mean_comment_disapproval	0.03	0.04	4.83
mean_comment_approval	0.08	0.08	2.29
mean_comment_admiration	0.06	0.11	4.14
mean_comment_disgust	0.00	0.01	17.68
mean_comment_anger	0.01	0.02	11.36



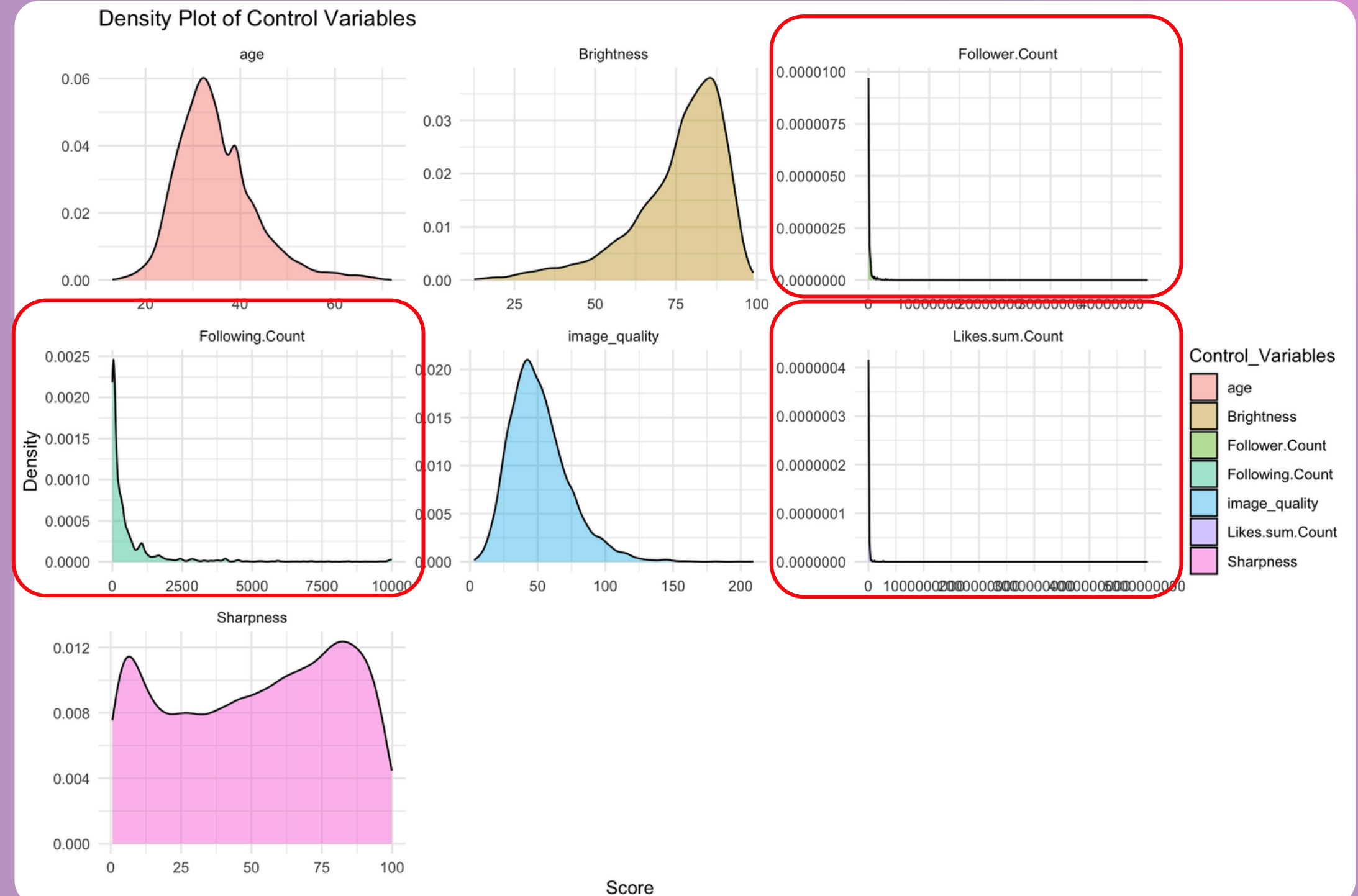
DESCRIPTIVE STATISTICS

Log-transforming Creator profile metrics such as:

- **Follower.Count**
- **Like.sum.Count**
- **Following.Count**

reduces skewness & improves distribution (skewness > 4)

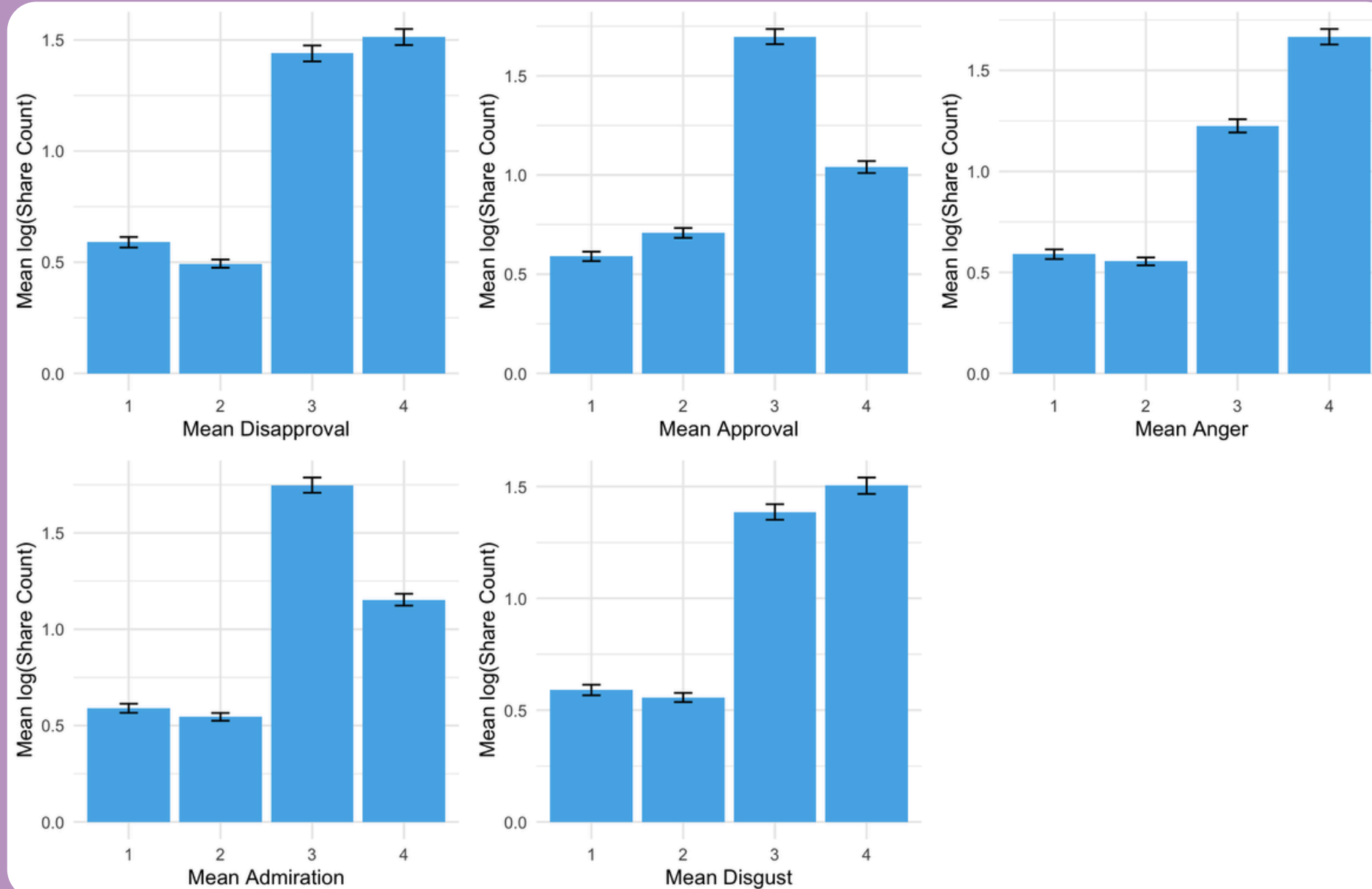
	mean	sd	skew
age	34.83	8.17	0.96
Follower.Count	213627.02	830695.40	28.32
Likes.sum.Count	7870690.45	66689284.83	62.42
Following.Count	660.55	1406.14	4.13
image_quality	51.42	22.11	1.07
Brightness	75.83	14.41	-1.32
Sharpness	51.06	29.92	-0.16



PRELIMINARY ANALYSIS

Initial analysis of how comment sentiments influence TikTok virality using summarySE

INITIAL INSIGHTS:



Negatively charged emotional comments—particularly those expressing disapproval, disgust, or anger—**are associated with higher virality (share count)**

Positively charged comments—like approval or admiration—have a **nonlinear impact on virality** δ may depend on context or interaction with other factors like creator popularity

SIMPLE OLS: DIAGNOSING MULTICOLLINEARITY

log_Share.Count ~ log_mean_disapproval + log_mean_approval + log_mean_admiration + log_mean_disgust + log_mean_anger + age + gender.deepface + race + log_Followers.Count + log_Likes.Count + log_Following.Count+ Verified.Status + image_quality + Brightness + Sharpness

Predictors	Coefficients (exp(β)-1)(%)	Interpretation
log_mean_disapproval	771.7 ***	A 1-unit increase in average disapproval in comments is associated with a 771.7% increase in share count, holding other variables constant .
log_mean_approval	46.8 (NS)	A 1-unit increase in average approval in comments is associated with a 46.8% increase in share count, but this effect is not statistically significant , so we cannot draw strong conclusions from it.
log_mean_anger	62.1 (NS)	A 1-unit increase in mean anger in comments is associated with a 62.1% increase in share count, but this effect is not statistically significant .
log_mean_admiration	-32.2 (NS)	A 1-unit increase in mean admiration in comments is associated with a 32.2% decrease in share count, but this effect is not statistically significant .
log_mean_disgust	-10.1 (NS)	A 1-unit increase in mean disgust in comments is associated with a 10.1% decrease in share count, but this effect is not statistically significant .

RIDGE REGRESSION

Adding L2 penalty to address multicollinearity

VIF:

log_followers 12.271950

log_likes 12.006602

Shrinks coefficients to prevent overfitting & handles multicollinearity between predictors followers and likes.

OPTIMAL LAMBDA: 0.077

Predictors	Coefficients (exp(β)-1) OLS Model (%)	Coefficients (exp(β)-1) Ridge Model (%)
log_mean_disapproval	771.7	724.7
log_mean_approval	46.8	47.5
log_mean_anger	62.1	61.4
log_mean_admiration	-32.2	-29.8
log_mean_disgust	-10.1	-7.97

VARIABLE SELECTION USING LASSO REGRESSION

Using Penalized Regression to Highlight Key Predictors

OPTIMAL LAMBDA: 0.0067

Variables Excluded:

- Mean_disgust
- Gender
- Race
- Verified Status
- Image Quality

Predictors	Coefficients ($\exp(\beta)-1$)(%)	Interpretation
log_mean_disapproval	707.4	Share (Virality) of TikTok video increases significantly with disapproval comments.
log_mean_approval	29.8	A strong and consistent positive predictor of shares. Approval comments also contribute to virality, though to a lesser extent than disapproval.
log_mean_anger	17.5	A strong and consistent positive predictor of shares.
log_mean_admiration	-27.5	Content receiving admiration appears to be shared less.
log_mean_disgust	0 (Excluded)	Not a strong predictor when other variables are included.

OLS POST-LASSO: REFINING OUR PREDICTORS

Used Lasso to identify key variables, then re-ran OLS to assess their impact more precisely

Key Predictors of TikTok Video Virality

- **Disapproval comments**
 - Most influential predictor
 - 772% increase in shares per unit increase in average disapproval comments
- **Anger comments**
 - Second strongest predictor
 - 51.2% increase in shares per unit increase in average anger comments

Negative emotional expressions—especially disapproval and anger—are strong drivers of share behavior, potentially fueling TikTok video virality.

Predictors	Coefficients ($\exp(\beta)-1$) Final OLS Model (%)
log_mean_disapproval	772 ***
log_mean_approval	45.7 (NS)
log_mean_anger	51.2 (NS)
log_mean_admiration	-34 .

How Does It Work Out of Sample?

for OLS Regression



Test Set (Train Set70 : Test Set30)

Metrics	Train Set	Test Set
R^2	0.2117	0.224
RMSE	1.5687	1.5137

- Compared with the training set, the test R^2 slightly improves & the test RMSE decreases
- Both are positive signs that the model generalizes well & does not overfit
- However, R^2 and RMSE still indicate only moderate predictive power, leaving room for improvement

How Does It Work Out of Sample?

for OLS Regression



K-Fold Cross Validation (K = 10)

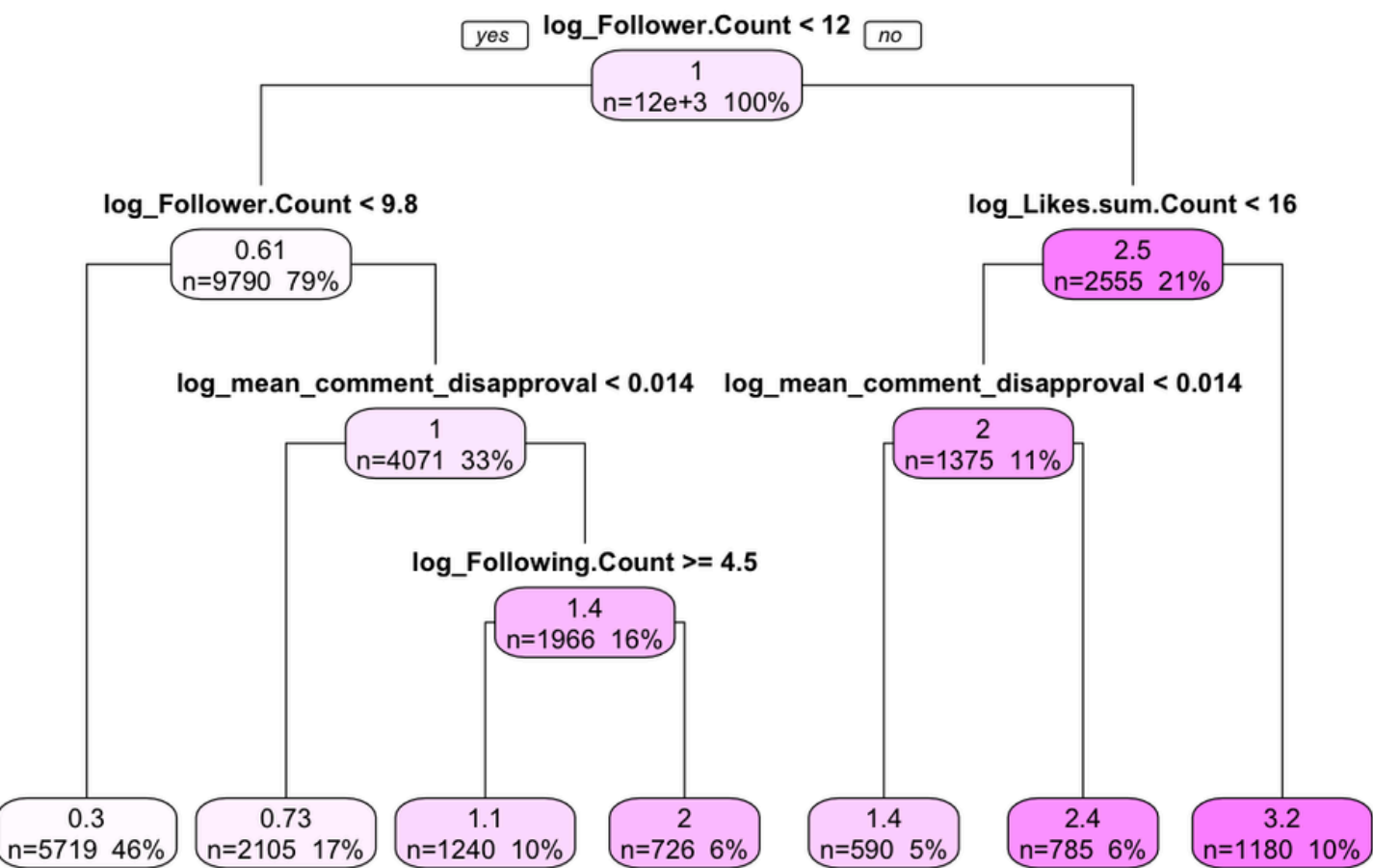
Metrics	Folds Average	Test Set
R ²	0.2148	0.2240
RMSE	1.5517	1.5137

- Why K-Fold CV: Repeatedly splits the dataset into training and validation folds, reducing bias from relying on a single random test split
- Provides a robustness check to confirm model generalizability
- The close alignment between 10-fold CV and test set metrics indicates model stability and low sensitivity to specific data splits

PREDICTING SHARE COUNT USING A REGRESSION TREE

Uncovering Key Drivers of Content Sharing Behavior

Regression Tree: Predicting TikTok Video Share Count



Variable	Importance	Variable	Importance
Follower Count	9308.760	Following Count	603.048
Like Count	7195.823	Mean Comment Anger	563.290
Mean Comment Disapproval	942.593	Mean Comment Admiration	481.246
Following Count	603.048	Mean Comment Approval	470.113

**all log-transformed*

WHAT THIS MEANS:

Creator popularity metrics (follower & like count) are the strongest predictors of share count

While impactful, comment disapproval is less predictive than creator popularity metrics

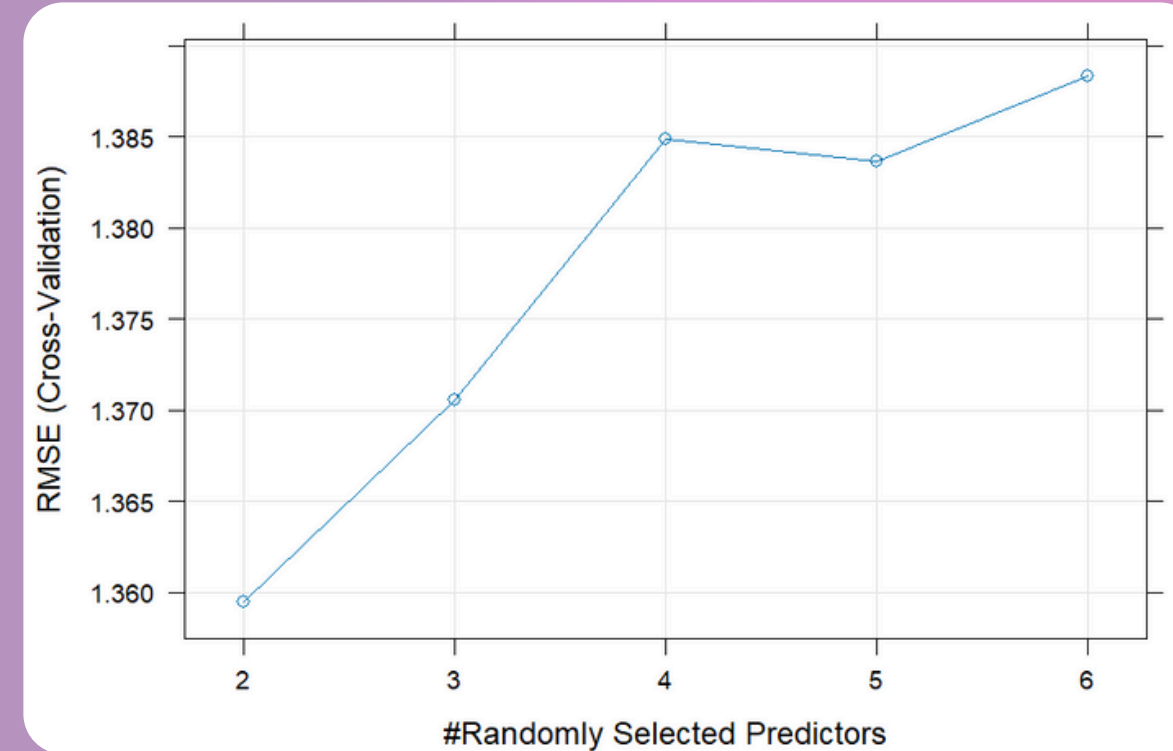
Remaining emotion-related comment metrics show modest importance in predicting share count

How Did We Tune the Random Forest Model?

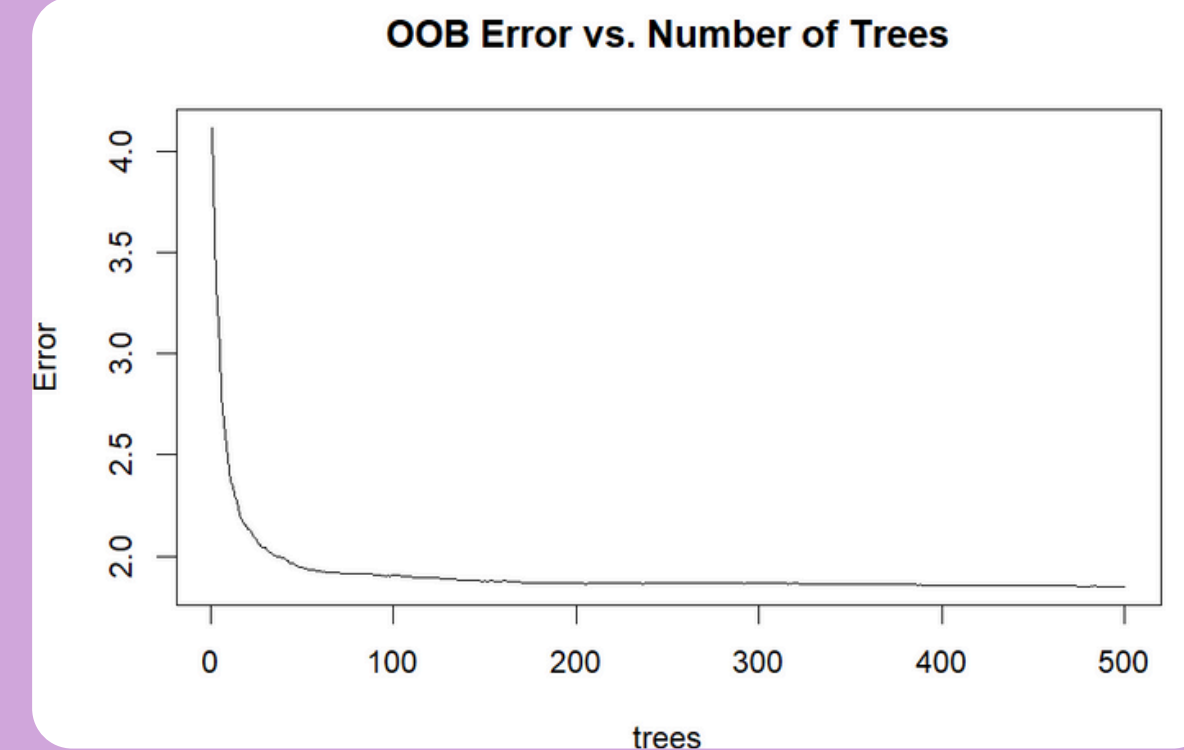
We set `ntree = 100` based on the OOB error curve plateauing, balancing stability and efficiency.

Using 10-fold cross-validation, we tuned `mtry` to minimize RMSE and enhance generalization

The final model was trained with these parameters to evaluate performance and generalizability



tune by having `mtry = 2`



`ntree = 100`

INSIGHTS FROM THE RANDOM FOREST MODEL

WHAT THIS MEANS:

Creator popularity metrics (follower, following, & like count) are the most critical in reducing error & increasing node purity

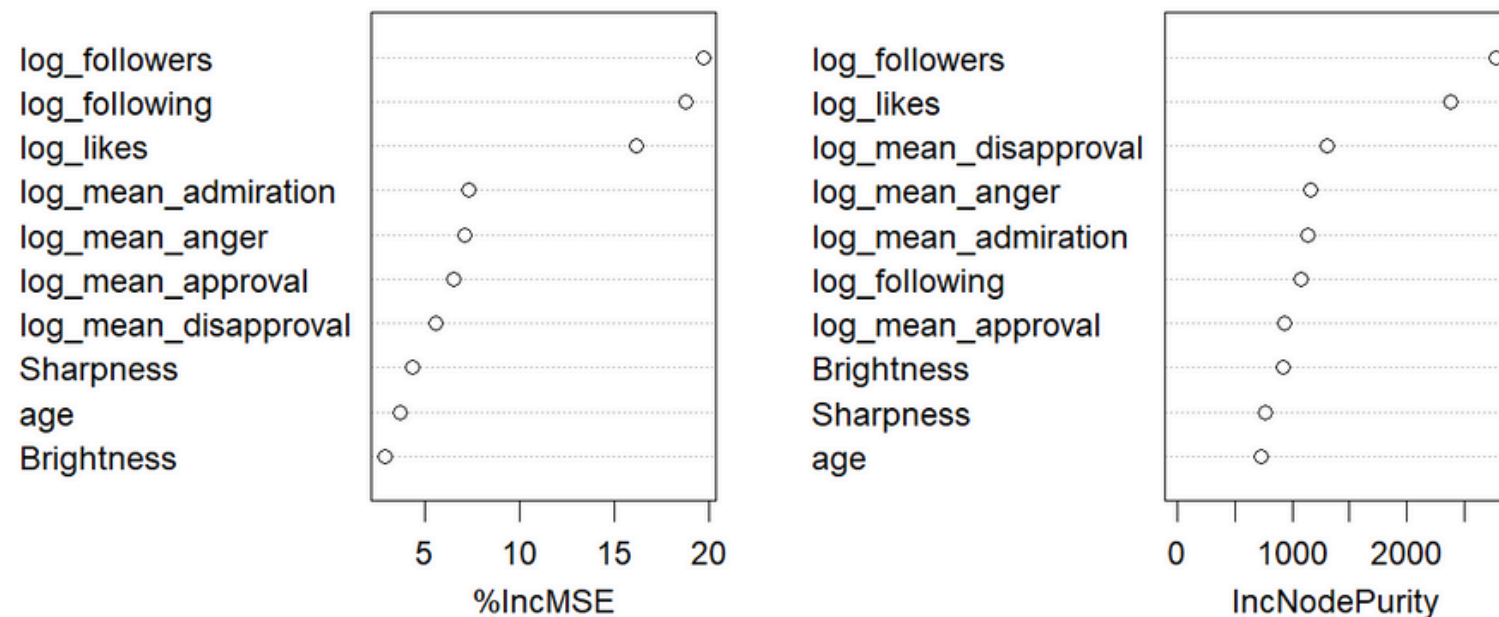
Comment sentiments still contribute to the model, but are secondary to popularity metrics

Age & video visual features (sharpness, brightness) are less influential in predicting share virality

RQ IMPLICATION FROM TREE-BASED MODELS:

Emotional expressions in comments, such as disapproval, **contribute to share virality but play a secondary role to creator popularity metrics**, amplifying share behavior only in certain contexts

Variable Importance (Final RF Model)



How Does It Work Out of Sample?

for Random Forest

Metrics	Train Set	Test Set	10 Folds Average
R ²	0.836	0.407	0.392
RMSE	0.705	1.376	1.359

- Good Predictive Power on Unseen Data
 - Our final Random Forest model performs well on the test set ($R^2 = 0.407$), showing **solid ability to predict real-world share behavior** beyond the training data.
- Cross-Validation Confirms Generalizability
 - The 10-fold CV metrics closely align with test set results, indicating strong generalization and minimal overfitting to any single data split.
- Train-Test Gap Reflects Memory Effect
 - The large gap between training ($R^2 = 0.836$) and test performance is expected in flexible models like Random Forest, which tend to memorize training data without validation constraints.

SUMMARY OF RESULTS

Models	Test RMSE	Test R ²	10-Fold CV RMSE	10-Fold CV R ²
OLS Regression	1.514	0.224	1.552	0.215
Random Forest	1.376	0.407	1.359	0.392

OLS Model

- **Higher disapproval sentiment → significantly higher share count** ($p < 0.001$)
- **Higher admiration sentiment → lower share count** ($p = 0.055$)
- Remaining comment sentiments → not statistically significant
- Creator popularity metrics (follower/ likes/ following count) → significant predictors of share count

Tree-Based Models

- Creator popularity metrics (follower / likes / following count) are most significant predictors of share virality
- **Average comment sentiments are secondary predictors of share virality**
 - **Negatively charged sentiments** (anger, disapproval) **have stronger predictive powers** than positively charged ones (approval, admiration)

MARKETING IMPLICATIONS

For TikTok Platform

Position TikTok as a Platform for Cultural Dialogue

- TikTok as a space for reaction, debate, & narrative-building
- **Emotionally charged content drives higher shares and fuels increased time on platform** + more stitches & duets
- Deep engagement boosts both algorithmic reach and ad revenue

Promote Constructive Engagement, Not Just Virality

- Prioritize content with lasting engagement signals, such as steady follower growth, audience engagement, growing sentiment over time
- These indicators reflect deeper audience connection beyond one-off virality

For TikTok Creators

Don't Shy Away from Polarizing Topics

Creating content that provokes strong emotional reactions attracts broader attention & shares;
controlled controversy can be a strategic growth tool

Balance Emotion and Reach

Emotional resonance can spark shareability, but a wide follower base sustains and amplifies it.

Avoid Overreliance on Shock

Don't overly rely on controversy for virality—while it may generate spikes, it may undermine long-term follower trust and hurt sustainable spreadability drivers

LIMITATIONS & FUTURE RESEARCH

Viewer's Real Emotion

- Some viewers may react emotionally but not leave a comment.
- The emotions detected in comments may not fully reflect how viewers actually felt while watching the video.



- Incorporate surveys or facial expression analysis tools to capture viewers' real-time emotions.

Emotion Detection Accuracy

- There might be a misclassification of emotion detection through the natural language analysis tools.



- Include more advanced tools to better capture sarcasm or other diverse languages.

Algorithmic Influence

- TikTok has a strong algorithm that shapes what viewers get or see.
- The higher the engagement among a specific group of users, the higher the possibility that the contents are shown or shared to them.



- Include other control variables such as the impressions-to-likes ratio to track the influence of algorithm.

REFERENCE

- Allsop Dee T., Bryce R. Bassett, and James A. Hoskins (2007), *Word-of-Mouth Research: Principles and Applications*, Journal of Advertising Research, 47 (4), 388–411.
- Berger, J., & Milkman, K. L. (2012). *What Makes Online Content Viral?* Journal of Marketing Research, 49 (2).
- Heath Chip, Bell Chris, and Sternberg Emily (2001), *Emotional Selection in Memes: The Case of Urban Legends*, Journal of Personality and Social Psychology, 81 (6), 1028–41.

CODE

```
library(tidyverse)
library(tidyr)
library(dplyr)
library(ggplot2)
library(psych)
```

```
tiktok <- tiktok %>% drop_na()
```

```
final_vars <- tiktok %>%
  select(mean_comment_disapproval, mean_comment_approval,
         mean_comment_admiration, mean_comment_disgust,
         mean_comment_anger, c, Like.Count,
         Age, gender.deepface, race, Follower.Count,
         Likes.sum.Count, Following.Count, Verified.Status,
         Image_quality, Brightness, Sharpness)
```

```
ggplot(final_vars, aes(x = log_share)) +
  geom_histogram(bins = 50, fill = "lightpink", color = "white") +
  labs(title = "Histogram of log1p(Share Count)",
       x = "log1p(Share Count)", y = "Frequency")
```

```
ggplot(final_vars, aes(x = Share.Count)) +
  geom_histogram(bins = 50, fill = "lightpink", color = "white") +
  labs(title = "Histogram of Share Count", x = "Share Count", y =
"Frequency")
```

```
options(scipen = 999)
numeric_vars <- final_vars %>%
  select(age, Follower.Count,
         Likes.sum.Count, Following.Count,
         Image_quality, Brightness, Sharpness)
```

```
describe(numeric_vars)
```

```
#numeric_vars skewness
nu_long <- numeric_vars %>%
  pivot_longer(everything(), names_to = "Control_Variables", values_to = "Value")
```

```
ggplot(nu_long, aes(x = Value, fill = Control_Variables)) +
  geom_density(alpha = 0.4) +
  facet_wrap(~ Control_Variables, scales = "free") +
  theme_minimal() +
  labs(title = "Density Plot of Control Variables", x = "Score", y = "Density")
```

```
#emotion_vars
emotion_vars <- final_vars %>%
  select(starts_with("mean_comment_"))
describe(emotion_vars)
```

```
emotion_long <- emotion_vars %>%
  pivot_longer(everything(), names_to = "Emotion", values_to = "Value")
```

```
#shape of each distribution
ggplot(emotion_long, aes(x = Value, fill = Emotion)) +
  geom_density(alpha = 0.4) +
  facet_wrap(~ Emotion, scales = "free") +
  theme_minimal() +
  labs(title = "Density Plot of Emotional Variables", x = "Score", y = "Density")
```

CODE

```
tiktok <- tiktok %>%
  mutate(
    gender.deepface = as.factor(gender.deepface),
    race             = as.factor(race),
    Verified.Status  = as.factor(Verified.Status)
  )

log_everything <- tiktok %>%
  mutate(
    log_share = log1p(Share.Count),
    log_followers = log1p(Follower.Count),
    log_likes = log1p(Likes.sum.Count),
    log_following = log1p(Following.Count),

    log_mean_disapproval =
log1p(mean_comment_disapproval),
    log_mean_approval    =
log1p(mean_comment_approval),
    log_mean_admiration  =
log1p(mean_comment_admiration),
    log_mean_disgust     =
log1p(mean_comment_disgust),
    log_mean_anger       = log1p(mean_comment_anger)
  ) %>%
  na.omit()
```

```
ols_log_everything_model <- lm(log_share ~
  log_mean_disapproval +
  log_mean_approval +
  log_mean_admiration +
  log_mean_disgust +
  log_mean_anger +
  age + gender.deepface + race +
  log_followers + log_likes + log_following +
  Verified.Status +
  image_quality + Brightness + Sharpness,
  data = log_everything)

summary(ols_log_everything_model)
odds_ratios <-
exp(coef(ols_log_everything_model))
percent_change <- (odds_ratios - 1) * 100

vif(ols_log_everything_model)
```

```
#Ridge
preds <- model.matrix(log_share ~
  log_mean_disapproval +
  log_mean_approval +
  log_mean_admiration +
  log_mean_disgust +
  log_mean_anger +
  age + gender.deepface + race + Verified.Status
+
  log_followers + log_likes + log_following +
  image_quality + Brightness + Sharpness,
  data = log_everything)[, -1]

outcome <- log_everything$log_share

cv_fit <- cv.glmnet(preds, outcome, alpha = 0) # alpha = 0
for Ridge

best_lambda_ridge <- cv_fit$lambda.min
print(best_lambda_ridge)

ridge_model <- glmnet(x = preds, y = outcome, alpha = 0,
  lambda = best_lambda_ridge)

coef(ridge_model)

odds_ratios <- exp(coef(ridge_model))
ridge_percent_change <- (odds_ratios - 1) * 100
```


CODE

```
#Lasso
set.seed(12345)

preds_lasso <- as.matrix(log_everything[, c(
  "log_mean_disapproval",
  "log_mean_approval",
  "log_mean_admiration",
  "log_mean_disgust",
  "log_mean_anger",
  "age", "gender.deepface", "race",
  "log_followers", "log_likes", "log_following", "Verified.Status",
  "image_quality", "Brightness", "Sharpness"
)])

outcome_lasso <- log_everything$log_share

cv_fit_lasso <- cv.glmnet(preds_lasso, outcome_lasso, alpha = 1)

best_lambda_lasso <- cv_fit_lasso$lambda.min
print(best_lambda_lasso)

#best lambda = 0.006723082

lasso_model <- glmnet(preds_lasso, outcome_lasso, alpha = 1,
lambda = 0.006723082)

coef(lasso_model)
odds_ratios <- exp(coef(lasso_model))
lasso_percent_change <- (odds_ratios -
1) * 100
```

```
Final OLS model:
log_everything <- tiktok %>%
  mutate(
    log_share = log1p(Share.Count),
    log_followers = log1p(Follower.Count),
    log_likes = log1p(Likes.sum.Count),
    log_following = log1p(Following.Count),

    log_mean_disapproval = log1p(mean_comment_disapproval),
    log_mean_approval = log1p(mean_comment_approval),
    log_mean_admiration = log1p(mean_comment_admiration),
    log_mean_disgust = log1p(mean_comment_disgust),
    log_mean_anger = log1p(mean_comment_anger)
  ) %>%
  na.omit()
ols_final_model <- lm(log_share ~
log_mean_disapproval +
log_mean_approval +
log_mean_admiration +
log_mean_anger +
age +
log_followers + log_likes + log_following +
Brightness + Sharpness,
data = log_everything)
summary(ols_final_model)
odds_ratios <- exp(coef(ols_final_model))
percent_change <- (odds_ratios - 1) * 100
```


CODE

(Regression Tree)

```
library(rpart)
library(rpart.plot)

set.seed(123)

# Fit regression tree
tree <- rpart(log_Share.Count ~ log_mean_comment_disapproval +
  log_mean_comment_approval + log_mean_comment_admiration
  + log_mean_comment_anger + age + log_Follower.Count +
  log_Likes.sum.Count + log_Following.Count +
  Brightness + Sharpness, data = tiktok,
  control = rpart.control(cp = 0.01,
    minsplit = 20,
    minbucket = 7))

summary(tree)
```

```
#rpart.plot(tree,
  type = 1,
  extra = 101,
  box.palette = "Purples",
  main = "Regression Tree: Predicting TikTok Video Share Count")

print(tree$variable.importance)

tree_sentiment <- rpart(log_Share.Count ~ log_mean_comment_disapproval +
  log_mean_comment_approval +
    log_mean_comment_admiration + log_mean_comment_disgust
  + log_mean_comment_anger,
  data = tiktok,
  control = rpart.control(cp = 0.01, minsplit = 20, minbucket = 7))

rpart.plot(tree_sentiment,
  type = 1,
  extra = 101,
  box.palette = "Blues",
  main = "Regression Tree: Predicting TikTok Video Share Count")

print(tree_sentiment$variable.importance)
```

CODE

(Random Forest)

```
install.packages("randomForest")
install.packages("caret")
install.packages("ggplot2")
library(randomForest)
library(caret)
library(ggplot2)
```

```
set.seed(12345)
```

```
vars_rf <- c("log_share",
            "log_mean_disapproval", "log_mean_approval",
            "log_mean_admiration", "log_mean_anger",
            "age", "log_followers", "log_likes", "log_following",
            "Brightness", "Sharpness")
```

```
train_idx <- sample(1:nrow(tiktok_log), 0.7 * nrow(tiktok_log))
train_data <- tiktok_log[train_idx, ]
test_data <- tiktok_log[-train_idx, ]
```

```
train_data <- train_data[complete.cases(train_data[, vars_rf]), ]
test_data <- test_data[complete.cases(test_data[, vars_rf]), ]
```

```
rf_model <- randomForest(
  log_share ~ .,
  data = train_data[, vars_rf],
  ntree = 500,
  mtry = 3, # sqrt(p) ≈ 3.3, testing 5
  importance = TRUE
)
```

```
plot(rf_model, main = "OOB Error vs. Number of Trees")
varImpPlot(rf_model, main = "Variable Importance")
```

```
#Tune mtry using 10-fold Cross-Validation
tune_grid <- expand.grid(.mtry = c(2, 3, 4, 5, 6))
```

```
final_rf <- randomForest(
  log_share ~ .,
  data = train_data[, vars_rf],
  ntree = 100,
  mtry = rf_tuned$bestTune$mtry,
  importance = TRUE
)
```

```
rf_tuned <- train(
  log_share ~ .,
  data = train_data[, vars_rf],
  method = "rf",
  tuneGrid = tune_grid,
  trControl = trainControl(method = "cv", number = 10),
  ntree = 100)
```

```
print(rf_tuned$bestTune)
```

```
final_rf <- randomForest(
  log_share ~ .,
  data = train_data[, vars_rf],
  ntree = 100,
  mtry = rf_tuned$bestTune$mtry,
  importance = TRUE)
varImpPlot(final_rf, main = "Variable Importance (Final
RF Model)")
```

```
pred_train <- predict(final_rf, newdata = train_data[, vars_rf])
pred_test <- predict(final_rf, newdata = test_data[, vars_rf])
```

```
rmse_train <- sqrt(mean((pred_train -
train_data$log_share)^2))
rmse_test <- sqrt(mean((pred_test - test_data$log_share)^2))
```

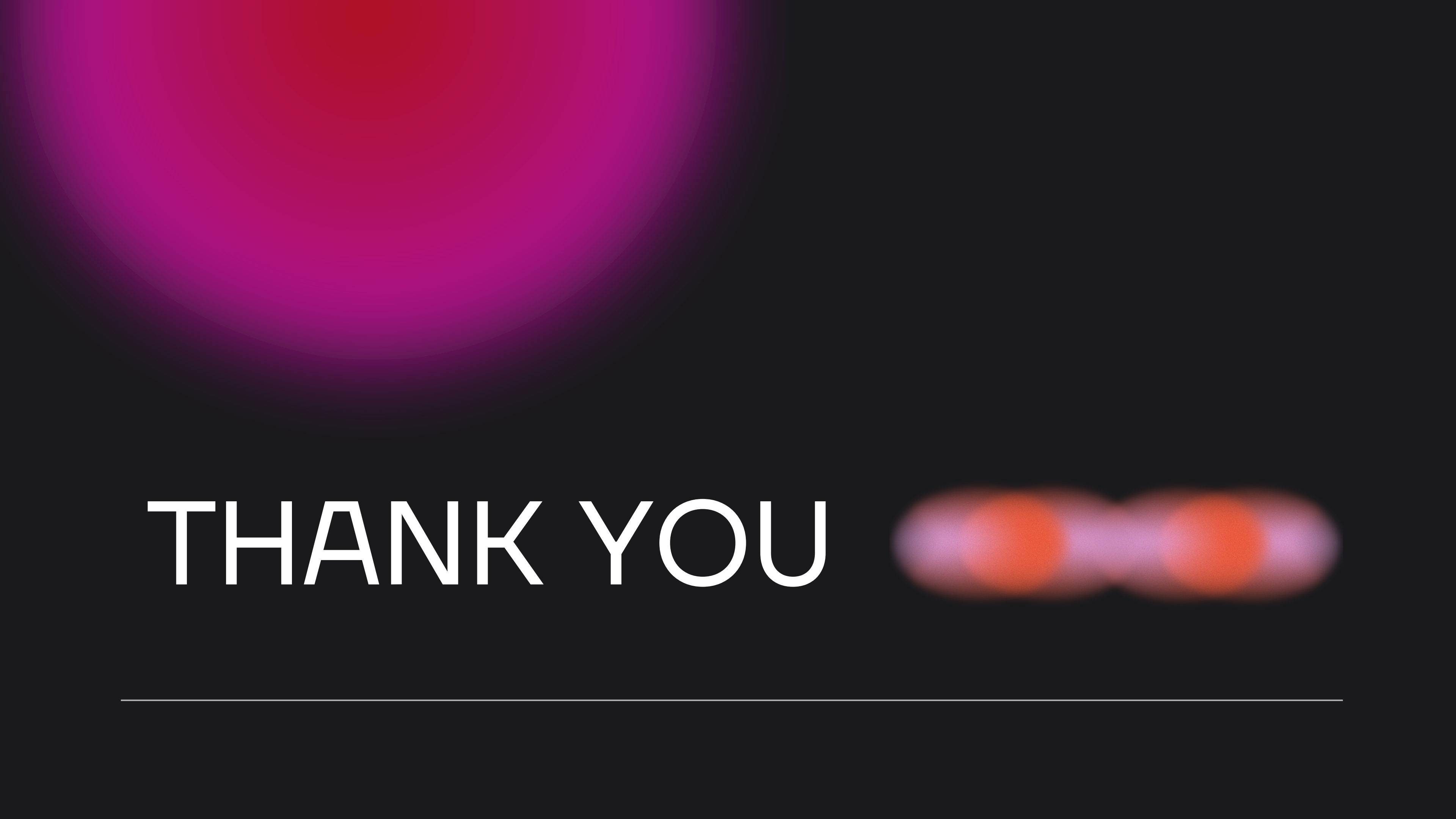
```
print(paste("Train RMSE:", round(rmse_train, 4)))
print(paste("Test RMSE:", round(rmse_test, 4)))
```

```
ss_total <- sum((df_pred$Actual - mean(df_pred$Actual))^2)
ss_resid <- sum((df_pred$Actual - df_pred$Predicted)^2)
r2_test <- 1 - ss_resid / ss_total
cat("Test Set R² (Random Forest):", round(r2_test, 4), "\n")
```

```
ss_total_train <- sum((train_data$log_share -
mean(train_data$log_share))^2)
ss_resid_train <- sum((train_data$log_share - pred_train)^2)
r2_train <- 1 - ss_resid_train / ss_total_train
cat("Train Set R²:", round(r2_train, 4), "\n")
```

```
cv_r2_values <- rf_tuned$resample$Rsquared
mean_cv_r2 <- mean(cv_r2_values)
cat("10-Fold Cross-Validated R² (Average):",
round(mean_cv_r2, 4), "\n")
```

```
cv_rmse_values <- rf_tuned$resample$RMSE
mean_cv_rmse <- mean(cv_rmse_values)
cat("10-Fold Cross-Validated RMSE (Average):",
round(mean_cv_rmse, 4), "\n")
```



THANK YOU
